



STEGANOGRAPHY

VS

CRYPTOGRAPHY

Hidden Security in the Digital World



CSA5101-Cryptography and Network Security

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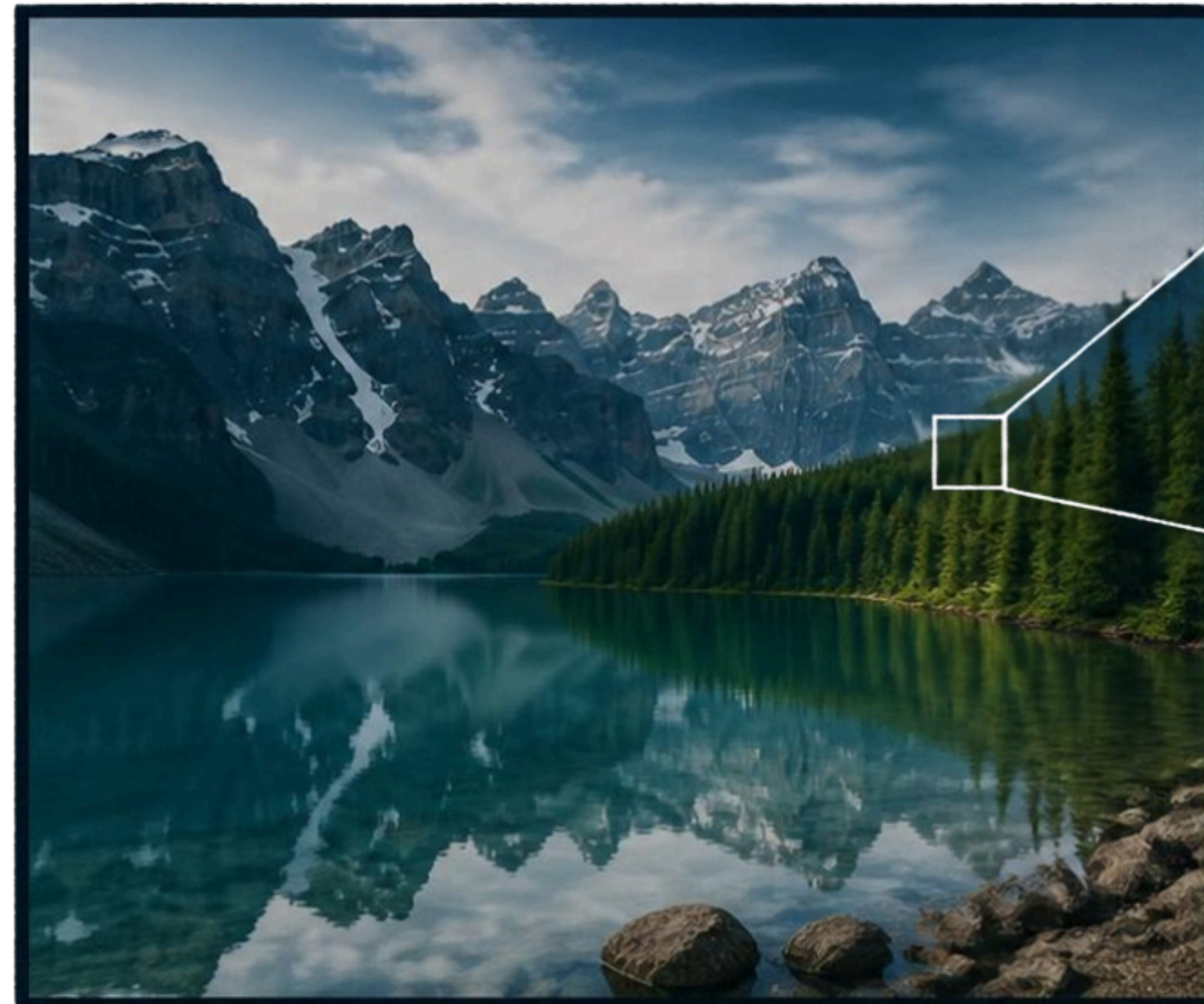
WHAT IS STEGANOGRAPHY?

Conceals information within
non-secret digital data

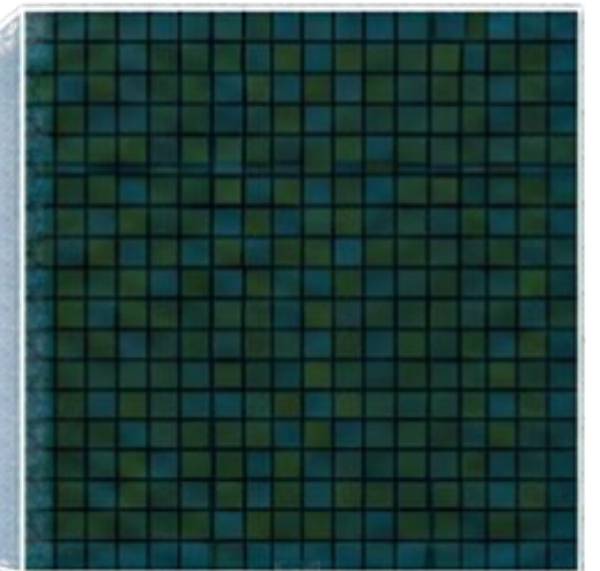
Hides the existence of
the message

Works with everyday media
like images, audio, and video

Example: embedding text
inside an image file



HIDDEN MESSAGE



MEET AT DAWN.
BRING THE FILE.

WHAT IS CRYPTOGRAPHY?

Transforms information into an unreadable format

Scrambles the content to protect confidentiality

Only authorized parties with the right key can restore it

Example: encrypted email or secure messaging



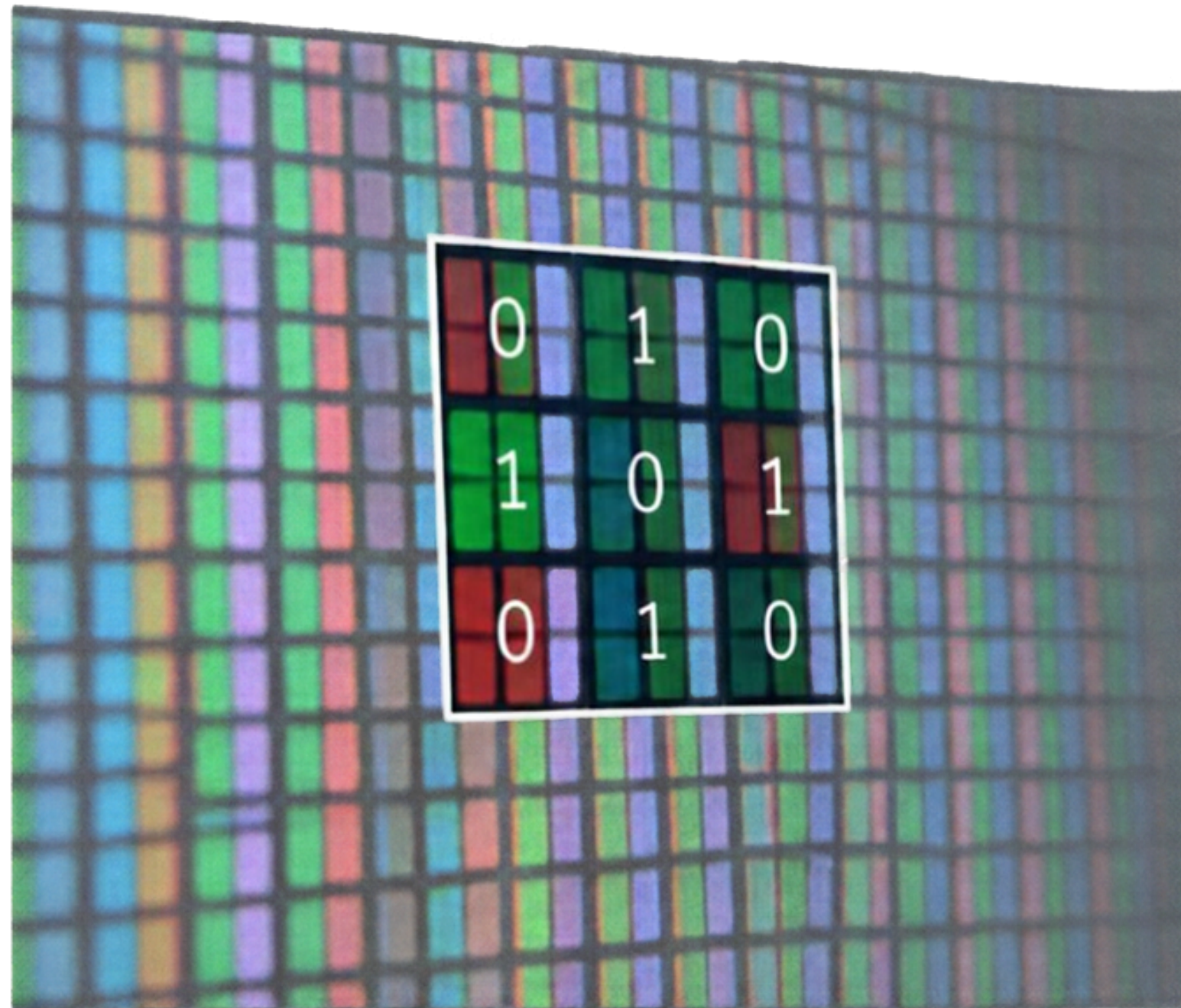
HOW STEGANOGRAPHY WORKS

Hides data inside ordinary files
so it is not visible to the eye

Alters the least significant bits
(LSB) of pixel values

Common covers: images,
audio, and video

The hidden data leaves no
visible trace



HIDDEN BITS

Data is stored by
changing the last
bit of each pixel
value.

HOW CRYPTOGRAPHY WORKS

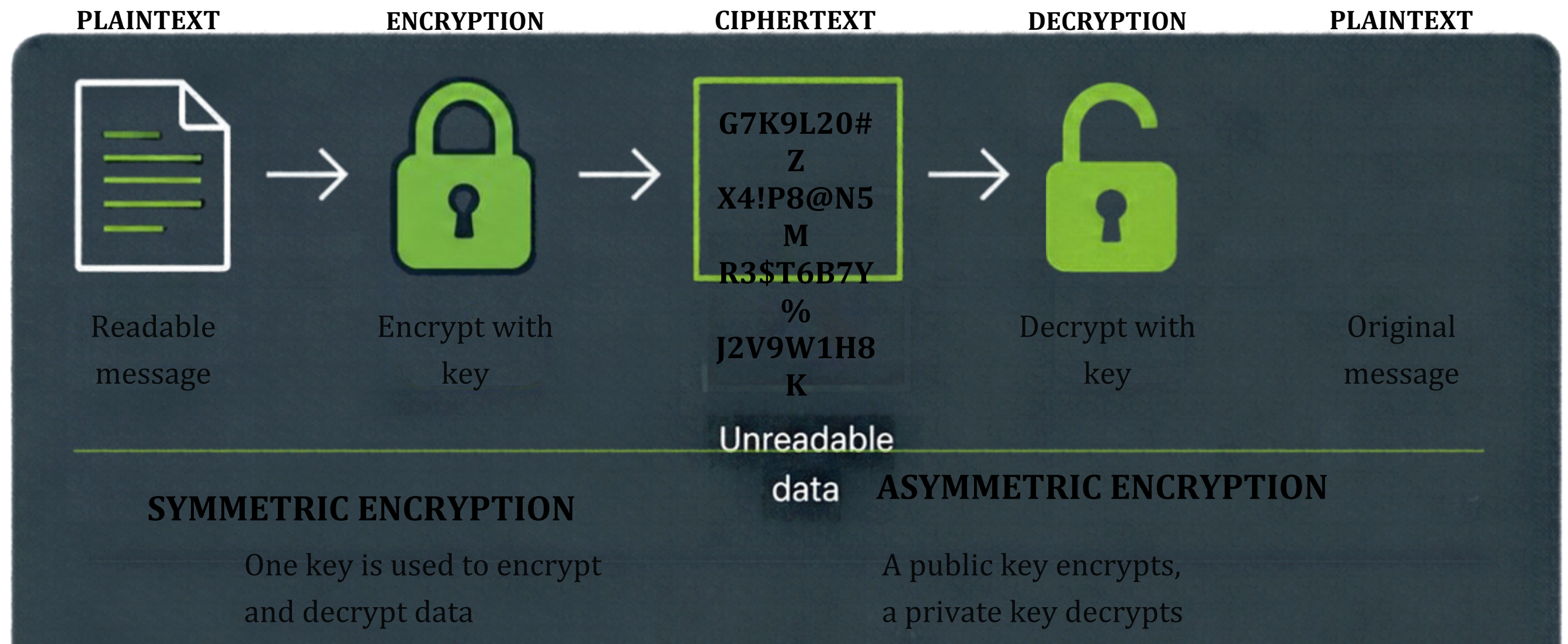


Protects data by transforming it into unreadable code

Symmetric and asymmetric encryption secure information

Common algorithms include AES and RSA

The right key restores the original message

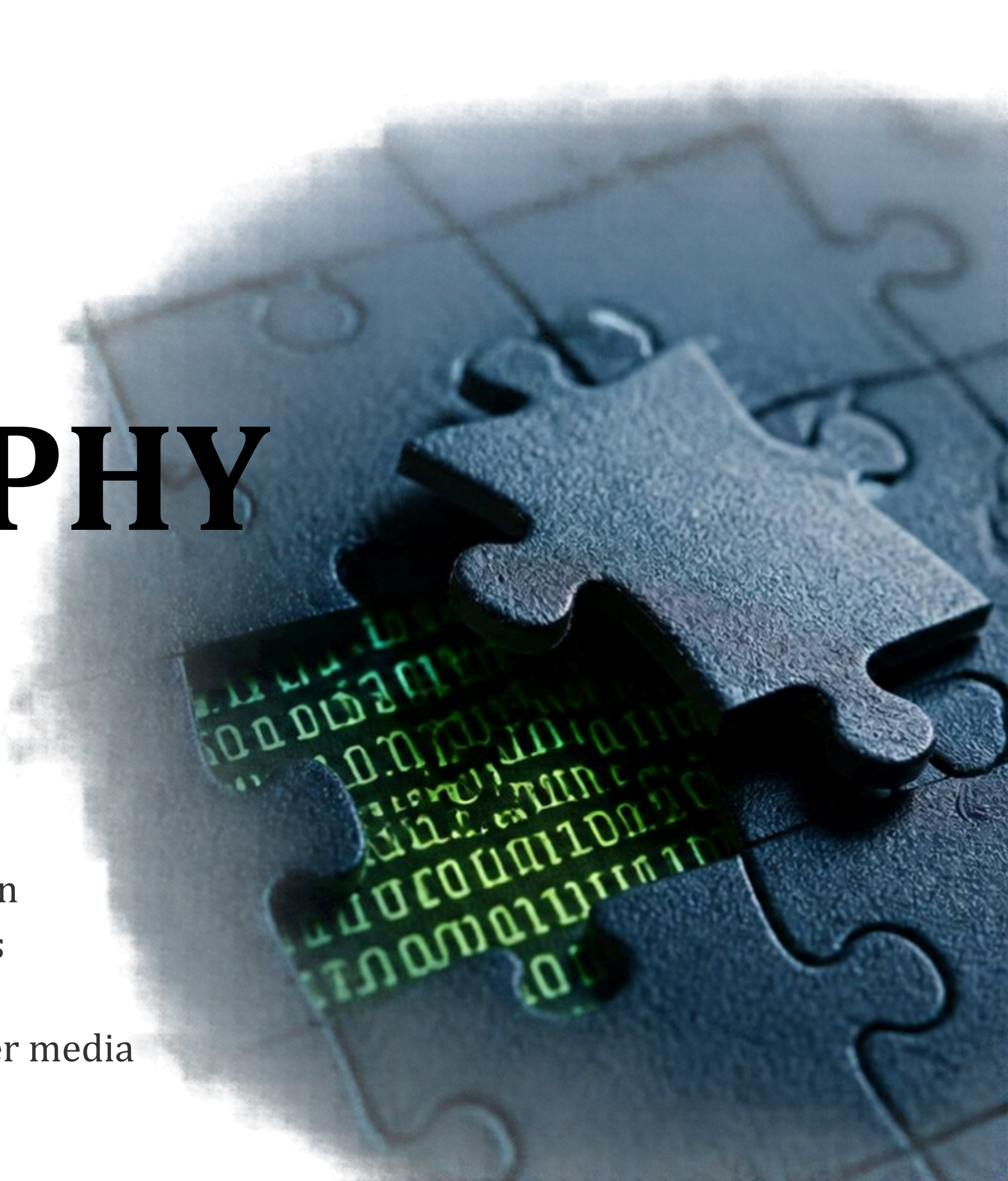


STEGANOGRAPHY VS CRYPTOGRAPHY

STEGANOGRAPHY	CRYPTOGRAPHY
Hides the existence of the message	Hides the content of the message
Used for covert communication	Used for secure communication
Can be vulnerable to steganalysis	Vulnerable if keys are compromised
Requires a cover medium	Does not require a cover medium



ADVANTAGES AND LIMITATIONS OF STEGANOGRAPHY



ADVANTAGES

Conceals the message's
existence

Can evade casual detection

LIMITATIONS

Limited data capacity

Vulnerable to detection
by specialized analysis

Requires suitable cover media

ADVANTAGES AND LIMITATIONS OF CRYPTOGRAPHY

ADVANTAGES

Strong confidentiality
and data integrity

Well-established
mathematical foundations

Widely used in
digital security

LIMITATIONS

Does not hide
message existence

Requires secure
key management

Can be computationally
intensive



PRACTICAL APPLICATIONS IN THE DIGITAL WORLD

Steganography

Embed and hide information
in plain sight

Cryptography

Protect data and ensure
trusted communication



SUMMARY AND NEXT STEPS

Steganography hides
the presence of information

Cryptography protects
the content of information

Both play important roles
in digital security

EXPLORE COMBINING BOTH FOR
ENHANCED SECURITY

